

# Wireless Communication Numerical Companion

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

## Ch 6 · Speech and Channel Coding Numerical Problems & Solutions

The examined calculation here is the **7-bit Hamming code** (80 Ch 5), worked both ways below. **Convolutional encoding, Viterbi decoding** (HOT 5/22) are asked as “explain”, but they are only convincing with a worked trellis, so one is given. The remaining problems are the tutorial-set coder-rate calculations, which are the standard follow-ups to the speech coding theory.

### The formula sheet

- **Hamming code**  $n = 2^m - 1$ ,  $k = 2^m - m - 1$ ,  $n - k = m$ ,  $d_{\min} = 3$ ,  $t = 1$ ,  $\text{rate} = k/n$
- **Detect  $t$  errors:**  $d' \geq t + 1$ . **Correct  $t$  errors:**  $d' \geq 2t + 1$
- **Convolutional code**  $\text{rate} = k/n$ , constraint length  $K$ ,  $2^{K-1}$  trellis states
- **Nyquist** a band-pass signal of bandwidth  $B$  needs  $f_s = 2B$  samples/s
- **Bit rate**  $R = f_s \times (\text{bits per sample})$
- **Quantizer**  $D = \int (x - f_Q(x))^2 p(x) dx$ ,  $S = E[x^2] = \int x^2 p(x) dx$ ,  $\text{SDR} = 10 \log_{10}(S/D)$
- **Coding gain**  $G = \left( \frac{E_b}{N_0} \right)_{\text{uncoded}} - \left( \frac{E_b}{N_0} \right)_{\text{coded}}$  dB, at the same BER

### Problem 1 7-bit Hamming code: encode and then detect

PIN 1/22 80 Ch 5 · the only pure channel-coding calculation in 22 papers, so it is worth banking exactly

QUESTION AS PRINTED 2080 Chaitra · Q7 · [3+5]

Explain the characteristics of Speech. Explain 7-bit hamming code method with relevant example.

The paper prints no data word, so supply your own. The run below encodes **1011** into a (7, 4) Hamming codeword, then receives a corrupted word and locates the faulty bit — that is the “relevant example” the 5 marks are for. **verified/added**

### Step 0: the structure, once

- For  $m = 3$ :  $n = 2^3 - 1 = 7$ ,  $k = 2^3 - 3 - 1 = 4$ , parity bits = 3.
- Parity bits go in positions that are **powers of 2**: 1, 2, 4. Data fills the rest: 3, 5, 6, 7.
- Layout, written with bit 7 on the *left* and bit 1 on the *right*:

$$D_7 \ D_6 \ D_5 \ P_4 \ D_3 \ P_2 \ P_1$$

- Each parity bit checks the positions whose binary index contains its own bit:

| Parity bit | Checks positions | Binary indices     | Why                |
|------------|------------------|--------------------|--------------------|
| $P_1$      | 1, 3, 5, 7       | 001, 011, 101, 111 | all have bit 0 set |
| $P_2$      | 2, 3, 6, 7       | 010, 011, 110, 111 | all have bit 1 set |
| $P_4$      | 4, 5, 6, 7       | 100, 101, 110, 111 | all have bit 2 set |

**This is the whole trick.** Because the groups are built on the binary indices, the three parity checks read together are the binary address of the faulty bit.

### Part A Encode the data word 1011 with even parity

- **Step 1: place the data bits into positions 7, 6, 5, 3:**

$$D_7 = 1, \quad D_6 = 0, \quad D_5 = 1, \quad D_3 = 1$$

- **Step 2: compute each parity bit so that its group has an even number of ones.**

| Parity | Group contents                          | Ones already present | Set to   |
|--------|-----------------------------------------|----------------------|----------|
| $P_1$  | $(P_1, D_3, D_5, D_7) = (P_1, 1, 1, 1)$ | 3 (odd)              | <b>1</b> |
| $P_2$  | $(P_2, D_3, D_6, D_7) = (P_2, 1, 0, 1)$ | 2 (even)             | <b>0</b> |
| $P_4$  | $(P_4, D_5, D_6, D_7) = (P_4, 1, 0, 1)$ | 2 (even)             | <b>0</b> |

- **Step 3: assemble the codeword, bit 7 first:**

|          |       |       |       |       |       |       |       |
|----------|-------|-------|-------|-------|-------|-------|-------|
| Position | 7     | 6     | 5     | 4     | 3     | 2     | 1     |
| Contents | $D_7$ | $D_6$ | $D_5$ | $P_4$ | $D_3$ | $P_2$ | $P_1$ |
| Value    | 1     | 0     | 1     | 0     | 1     | 0     | 1     |

transmitted codeword = 1010101

**Part B** A codeword 1011011 is received. Is it correct? If not, locate the error.

- **Step 1: lay out the received bits.**

|          |   |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|---|
| Position | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| Received | 1 | 0 | 1 | 1 | 0 | 1 | 1 |

- **Step 2: recheck each parity group, still assuming even parity.**

| Check | Positions  | Bits read  | Parity      | Syndrome bit |
|-------|------------|------------|-------------|--------------|
| $P_1$ | 1, 3, 5, 7 | 1, 0, 1, 1 | ODD → error | 1            |
| $P_2$ | 2, 3, 6, 7 | 1, 0, 0, 1 | EVEN → ok   | 0            |
| $P_4$ | 4, 5, 6, 7 | 1, 1, 0, 1 | ODD → error | 1            |

- **Step 3: read the syndrome.** Write it as  $P_4P_2P_1$ :

$$\text{syndrome} = 101_{(2)} = 4 + 0 + 1 = 5$$

- **Step 4: conclude.**

The received word is INCORRECT; bit position 5 is in error.

- **Step 5: correct it by flipping bit 5, 1 → 0:**

$$\text{corrected codeword} = 1001011$$

$$\text{recovered data (positions 7,6,5,3)} = 1000$$

- **Non-zero syndrome = error, and its decimal value is the position. A zero syndrome means no single-bit error.**
- **Limitation to state:**  $d_{\min} = 3$ , so this code corrects **one** bit and detects **two**. A **two-bit** error produces a non-zero syndrome pointing at the *wrong* position, and “correcting” it makes things worse. **That is why Hamming codes are always paired with interleaving** (Ch 5) on a fading channel, where errors arrive in bursts.

**Hamming code parameters for other  $m$**

| $m$ | $n = 2^m - 1$ | $k = 2^m - m - 1$ | parity bits | rate $k/n$ |
|-----|---------------|-------------------|-------------|------------|
| 3   | 7             | 4                 | 3           | 0.571      |
| 4   | 15            | 11                | 4           | 0.733      |
| 5   | 31            | 26                | 5           | 0.839      |
| 6   | 63            | 57                | 6           | 0.905      |

The rate improves as  $m$  grows, but the code still corrects only **one** bit per block, so the protection per bit gets weaker. That trade-off is worth one line in any Hamming answer.

## Problem 2 Convolutional encoding and Viterbi decoding

**HOT 5/22** 74 Bh **5**, 75 Bh, 70 Ma, 72 Ash **3**, 73 Bh **5**

**QUESTION AS PRINTED** 5 papers, always as a short note

Write short notes on: **Convolutional encoding and decoding** (2074 Bhadra Q9 c **5**)

Write short notes on: **Viterbi decoding algorithm** (2073 Bhadra Q9 c **5** · 2072 Ashwin Q9 c **3** · 2075 Bhadra Q12 b · 2070 Magh Q10 c)

**PROBLEM** not a PYQ · the example that carries the short note verified/added

For the rate  $1/2$ , constraint length  $K = 3$  convolutional encoder defined by  $X_1 = M + M_1 + M_2$  and  $X_2 = M + M_2$  (mod-2 addition), encode the message **1011**. Then decode the received sequence  $Y = 11\ 01\ 10$  using the Viterbi algorithm, showing the trellis and the surviving path metrics.

### Part A Encoding 1011

- State = the contents of the two memory stages ( $M_1M_2$ ), so there are  $2^{K-1} = 4$  states:  $a = 00, b = 01, c = 10, d = 11$ .
- Two flush (tail) bits are appended so the encoder returns to state  $a$ .

| Input $M$ | State ( $M_1M_2$ ) | $X_1 = M + M_1 + M_2$ | $X_2 = M + M_2$ | Output    | Next state |
|-----------|--------------------|-----------------------|-----------------|-----------|------------|
| 1         | 00                 | $1+0+0 = 1$           | $1+0 = 1$       | <b>11</b> | 10         |
| 0         | 10                 | $0+1+0 = 1$           | $0+0 = 0$       | <b>10</b> | 01         |
| 1         | 01                 | $1+0+1 = 0$           | $1+1 = 0$       | <b>00</b> | 10         |
| 1         | 10                 | $1+1+0 = 0$           | $1+0 = 1$       | <b>01</b> | 11         |
| 0         | 11                 | $0+1+1 = 0$           | $0+1 = 1$       | <b>01</b> | 01         |
| 0         | 01                 | $0+0+1 = 1$           | $0+1 = 1$       | <b>11</b> | 00         |

encoded output = 11 10 00 01 01 11

- 4 message bits plus 2 flush bits give 6 input bits, and at rate  $1/2$  that is **12 output bits**. ✓
- **Code rate including the tail** is  $4/12 = 1/3$ , not  $1/2$ . Real systems use long frames so the tail overhead is negligible. Worth a line.

### Part B Viterbi decoding of $Y = 11\ 01\ 10$

- **Metric used: Hamming distance** between the branch's expected output and the received pair. Lower is better.
- At every node keep only the **smaller-metric** incoming path (the survivor) and discard the other.

| Step | Received | State | Path metric | Surviving input path |
|------|----------|-------|-------------|----------------------|
| 1    | 11       | 00    | 2           | 0                    |
|      |          | 10    | <b>0</b>    | 1                    |
| 2    | 01       | 00    | 3           | 0, 0                 |
|      |          | 01    | 2           | 1, 0                 |
|      |          | 10    | 3           | 0, 1                 |
|      |          | 11    | <b>0</b>    | 1, 1                 |
| 3    | 10       | 00    | 3           | 1, 0, 0              |
|      |          | 01    | 2           | 1, 1, 0              |
|      |          | 10    | 3           | 1, 0, 1              |
|      |          | 11    | <b>0</b>    | 1, 1, 1              |

### Reading the table

- **Step 1.** From state 00, input 0 gives output 00 (distance 2 from the received 11), input 1 gives output 11 (distance 0). So state 10 carries metric 0.
- **Step 2.** From state 10, input 0 gives 10 (distance 1 from 01) and input 1 gives 01 (distance 0). The path 1, 1 into state 11 keeps metric 0.
- **Step 3.** From state 11, input 1 gives output 10, which **matches the received 10 exactly**, so the metric stays 0.
- **Survivor:** the path ending in state 11 with total metric 0.

decoded message = 1 1 1

- **Verify by re-encoding** 1, 1, 1 from state 00: 11, then 01, then 10 =  $Y$  exactly. ✓ **Metric 0 means the received sequence was a valid codeword, ie no channel errors.**
- **Why Viterbi is tractable:** after every step there are still only 4 survivors, never more, because each state keeps one path. An exhaustive tree search would have  $2^L$  paths after  $L$  bits.

### Problem 3 Sub-band coder bit rate

PIN 1/22 80 Bb 6 · the calculation behind “sub-band coding with transmitter and receiver”

QUESTION AS PRINTED 2080 Bhadra · Q6 (a) · [6]

Explain sub-band coding with transmitter and receiver.

PROBLEM not a PYQ · the rate sum behind the block diagram verified/added

A sub-band coder splits the input into the four sub-bands tabulated below and samples each at its band-pass Nyquist rate. Find the total encoding bit rate, and compare it with coding the full band at the same resolution.

| Sub-band | Band (Hz) | Bandwidth | $f_s = 2B$ | Bits/sample | Rate (bps) |
|----------|-----------|-----------|------------|-------------|------------|
| 1        | 225–450   | 225 Hz    | 450        | 4           | 1800       |
| 2        | 450–900   | 450 Hz    | 900        | 3           | 2700       |
| 3        | 1000–1500 | 500 Hz    | 1000       | 2           | 2000       |
| 4        | 1800–2700 | 900 Hz    | 1800       | 1           | 1800       |
| Total    |           |           |            |             | 8300       |

- **Step 1: bandwidth of each band**  $= f_{\text{high}} - f_{\text{low}}$ .
- **Step 2: sampling rate.** For **perfect reconstruction** each band-pass signal must be sampled at **twice its own bandwidth**, not twice its highest frequency. **That saving is the whole reason for the low-pass translation step in the sub-band coder.**

$$f_{s,1} = 2(450 - 225) = 450 \text{ samples/s}, \quad f_{s,2} = 2(900 - 450) = 900$$

$$f_{s,3} = 2(1500 - 1000) = 1000, \quad f_{s,4} = 2(2700 - 1800) = 1800$$

- **Step 3: total rate.**

$$R = 450(4) + 900(3) + 1000(2) + 1800(1)$$

$$= 1800 + 2700 + 2000 + 1800 = \boxed{8300 \text{ bps} = 8.3 \text{ kbps}}$$

- **Note the bit allocation:** the **narrowest, lowest** band gets the **most** bits (4), and the widest, highest band gets the fewest (1). That follows directly from the **PSD property** of speech: low bands carry most of the energy and most of the perceptual weight.
- **Compare:** plain 8 kHz / 8-bit PCM of the same speech would need **64 kbps**. Sub-band coding achieves nearly **8:1 compression** while remaining a waveform coder.

### Problem 4 Upper bound on speech coder rate in an FDMA system

tutorial set; the calculation behind “choosing a speech codec for mobile communications”

PROBLEM not a PYQ · the sum behind “choosing a speech codec” verified/added

A digital mobile system uses 810–826 MHz for the forward channel and 940–956 MHz for the reverse channel. 90% of the bandwidth is given to traffic channels, and the system must support at least 1150 simultaneous calls using FDMA. The modulation achieves a spectral efficiency of 1.68 bps/Hz, and channel impairments force rate 1/2 forward error correction. Find the upper bound on the transmission bit rate that a speech coder in this system can use.

- **Step 1: total band.**

$$B_{\text{total}} = 826 - 810 = 16 \text{ MHz}$$

The reverse band is the same width, and FDMA pairs them, so work with one direction.

- **Step 2: bandwidth actually available to traffic.**

$$B_{\text{traffic}} = 0.9 \times 16 = 14.4 \text{ MHz}$$

**The other 10 % goes to control channels and guard bands.**

- **Step 3: bandwidth per channel.** FDMA divides it among the simultaneous calls:

$$B_{\text{ch}} = \frac{14.4 \text{ MHz}}{1150} = 12.52 \text{ kHz}$$

$$\Rightarrow \boxed{\text{maximum channel bandwidth} = 12.5 \text{ kHz}}$$

- **Step 4: raw bit rate the channel can carry.**

$$R_{\text{raw}} = 1.68 \frac{\text{bps}}{\text{Hz}} \times 12.52 \text{ kHz} = 21.04 \text{ kbps}$$

- **Step 5: subtract the FEC overhead.** A rate 1/2 code doubles the bits, so only half the raw rate is available to the speech coder:

$$R_{\text{speech}} = 0.5 \times 21.04 = \boxed{10.52 \text{ kbps}}$$

- **Interpretation:** the speech coder must run at **10.5 kbps or below**. A plain waveform coder cannot do that at toll quality, so this system needs a **hybrid or vocoder** class coder, eg RPE-LTP at 13 kbps (too fast here) or a CELP variant at 8 kbps (fits).
- **This is exactly the reasoning that forces cellular systems onto low-rate coders**, and it is the numerical answer to “why do we need speech coding techniques”.

### Problem 5 Quantizer mean square error and signal to distortion ratio

tutorial set; the calculation behind the PDF property of speech **TOP 7/22**

**PROBLEM** not a PYQ · the PDF property of speech, made numerical **verified/added**

The input signal to a quantizer has the probability density function  $p(x) = x/32$  for  $0 \leq x \leq 8$  and zero elsewhere. The quantization levels are  $\{1, 3, 5, 7\}$ .

- Compute the mean square error distortion at the quantizer output.
- Compute the output signal to distortion ratio.
- How would you redistribute the quantization levels to decrease the distortion, and for what input pdf would this quantizer already be optimal?

- **Step 0: check the pdf is valid.**

$$\int_0^8 \frac{x}{32} dx = \left[ \frac{x^2}{64} \right]_0^8 = \frac{64}{64} = 1 \quad \checkmark$$

- **Step 1: set the decision boundaries.** With levels at 1, 3, 5, 7 the midpoints give boundaries at  $\{0, 2, 4, 6, 8\}$ , so the quantizer is **uniform** with step 2.
- **Step 2: mean square error distortion.**

$$\begin{aligned} D &= \int_{-\infty}^{\infty} (x - f_Q(x))^2 p(x) dx \\ &= \int_0^2 (x - 1)^2 \frac{x}{32} dx + \int_2^4 (x - 3)^2 \frac{x}{32} dx + \int_4^6 (x - 5)^2 \frac{x}{32} dx + \int_6^8 (x - 7)^2 \frac{x}{32} dx \end{aligned}$$

- **Step 3: evaluate one integral to show the method.** For the first,

$$\begin{aligned} \int_0^2 (x^2 - 2x + 1) \frac{x}{32} dx &= \frac{1}{32} \int_0^2 (x^3 - 2x^2 + x) dx \\ &= \frac{1}{32} \left[ \frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} \right]_0^2 = \frac{1}{32} \left( 4 - \frac{16}{3} + 2 \right) = \frac{1}{32} \cdot \frac{2}{3} = \frac{1}{48} \end{aligned}$$

The other three evaluate to  $\frac{3}{48}$ ,  $\frac{5}{48}$  and  $\frac{7}{48}$  by the same method.

- **Step 4: total distortion.**

$$D = \frac{1 + 3 + 5 + 7}{48} = \frac{16}{48} = \frac{1}{3} = \boxed{0.3333}$$

- **Step 5: signal power.**

$$S = E[x^2] = \int_0^8 x^2 \frac{x}{32} dx = \frac{1}{32} \left[ \frac{x^4}{4} \right]_0^8 = \frac{1}{32} \cdot \frac{4096}{4} = \boxed{32}$$

- **Step 6: signal to distortion ratio.**

$$\text{SDR} = 10 \log_{10} \left( \frac{S}{D} \right) = 10 \log_{10} \left( \frac{32}{0.3333} \right) = 10 \log_{10}(96.0) = \boxed{19.82 \text{ dB}}$$

- **Step 7: how to reduce the distortion.**

- Note that the four sub-integrals came out as 1, 3, 5, 7 over 48: **the top interval contributes seven times as much distortion as the bottom one.**
- That is because  $p(x) = x/32$  rises with  $x$ , so most of the probability mass sits at large amplitudes.
- **Fix: use a non-uniform quantizer.** Place **more levels where the probability is high** (large  $x$ ) and fewer where it is low (small  $x$ ), ie shrink the step size towards  $x = 8$ .
- This is exactly the **PDF property** of speech being exploited, and it is the principle behind  $\mu$ -law and A-law companding.
- **Step 8: for what pdf is this quantizer optimal?** A uniform quantizer is optimal when every interval contributes equally, which happens only when the input is **uniformly distributed** over  $[0, 8]$ , ie  $p(x) = 1/8$ .

### Problem 6 Speech coder rates: what fits in what

supporting numbers for the vocoder **TOP 10/22** and LPC **TOP 8/22** questions

Reference figures. **Quoting one correct rate is often worth a mark** in an otherwise descriptive answer.

| Coder                    | Class    | Bit rate                          | Note                       |
|--------------------------|----------|-----------------------------------|----------------------------|
| PCM (toll reference)     | Waveform | 64 kbps                           | 8 kHz, 8 bit, $\mu$ -law   |
| ADPCM                    | Waveform | 32 kbps                           | G.726                      |
| Sub-band coding          | Waveform | 9.6–32 kbps                       | 4 or 8 bands               |
| Adaptive transform (ATC) | Waveform | 9.6–20 kbps                       | DCT based                  |
| Voice-excited vocoder    | Hybrid   | 7.2–9.6 kbps                      | no pitch extraction needed |
| GSM RPE-LTP              | Hybrid   | <b>13 kbps</b>                    | full rate GSM              |
| CELP                     | Vocoder  | <b>4.8 kbps</b>                   | codebook excitation        |
| LPC                      | Vocoder  | 4.8 kbps (good quality), 2.4 kbps | all-pole model             |

- **Compression achieved** against the 64 kbps PCM reference: ADPCM  $2\times$ , sub-band  $\approx 7\times$ , GSM  $\approx 5\times$ , CELP  $\approx 13\times$ .
- **The pattern to remember:** rate falls, complexity rises, and robustness falls. A 4.8 kbps CELP frame damaged by the channel is far more audible than a damaged 64 kbps PCM sample. **That is why the lowest-rate coders need the strongest channel coding**, which is the link between the two halves of this chapter.

### Problem 7 Coding gain

**syllabus, no PYQ yet** · included because it is the number that justifies every code in the chapter

**PROBLEM** not a PYQ · the number that justifies every code here **verified/added**

An uncoded BPSK link needs  $E_b/N_0 = 10.5$  dB to reach a BER of  $10^{-5}$ . With a rate  $1/2$ ,  $K = 7$  convolutional code and soft-decision Viterbi decoding, the same BER is reached at  $E_b/N_0 = 4.5$  dB. Find the coding gain in dB and the transmit power saving it represents.

- **Step 1: coding gain.**

$$G = \left( \frac{E_b}{N_0} \right)_{\text{uncoded}} - \left( \frac{E_b}{N_0} \right)_{\text{coded}} = 10.5 - 4.5 = \boxed{6 \text{ dB}}$$

- **Step 2: what 6 dB is worth in power.**

$$10^{6/10} = 3.98 \approx 4$$

**The transmitter can run at one quarter of the power for the same quality.**

- **Step 3: what it is worth in range**, at path loss exponent  $n = 3.5$ :

$$10^{6/(10 \times 3.5)} = 10^{0.1714} = 1.48$$

ie **48 % more range**, or  $1.48^2 = 2.2$  times the area from one site.

- **Step 4: what it costs.** The rate  $1/2$  code **doubles the transmitted bit rate**, so it needs twice the bandwidth (or halves the user data rate).
- **TCM exists precisely to escape that cost:** by combining coding with a larger signal constellation it delivers coding gains up to 6 dB with **no bandwidth expansion**. Turbo codes then push the gain further still, close to the Shannon bound.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                                       | Short    | Means                                      |
|----------|---------------------------------------------|----------|--------------------------------------------|
| BS       | base station                                | BTS      | base transceiver station                   |
| MS       | mobile station                              | MSC      | mobile switching centre                    |
| BSC      | base station controller                     | MSK      | minimum shift keying                       |
| CCI      | co-channel interference                     | ACI      | adjacent channel interference              |
| SIR, S/I | signal to interference ratio                | SNR, S/N | signal to noise ratio                      |
| GoS      | grade of service                            | MAHO     | mobile assisted handoff                    |
| FCA      | fixed channel assignment                    | DCA      | dynamic channel assignment                 |
| LOS      | line of sight                               | NLOS     | non line of sight                          |
| FDMA     | frequency division multiple access          | TDMA     | time division multiple access              |
| CDMA     | code division multiple access               | OFDM     | orthogonal frequency division multiplexing |
| FDD      | frequency division duplex                   | TDD      | time division duplex                       |
| GSM      | global system for mobile communications     | AMPS     | advanced mobile phone system               |
| PCS      | personal communication system               | DAS      | distributed antenna system                 |
| BCCH     | broadcast control channel                   | EIRP     | effective isotropic radiated power         |
| ERP      | effective radiated power                    | RF       | radio frequency                            |
| Tx, TX   | transmitter                                 | Rx, RX   | receiver                                   |
| ISI      | intersymbol interference                    | PDP      | power delay profile                        |
| DFE      | decision feedback equalizer                 | MLSE     | maximum likelihood sequence estimation     |
| LMS      | least mean square                           | RLS      | recursive least squares                    |
| MSE      | mean square error                           | MMSE     | minimum mean square error                  |
| ZF       | zero forcing                                | MRC      | maximal ratio combining                    |
| EGC      | equal gain combining                        | BER      | bit error rate                             |
| PCM      | pulse code modulation                       | DPCM     | differential PCM                           |
| ADPCM    | adaptive differential PCM                   | LPC      | linear predictive coder                    |
| RPE-LTP  | regular pulse excited, long term prediction | CELP     | code excited linear prediction             |
| RELTP    | residual excited linear prediction          | MPE      | multipulse excited                         |
| VSELP    | vector sum excited linear prediction        | ATC      | adaptive transform coding                  |
| SBC      | sub-band coding                             | DCT      | discrete cosine transform                  |
| FEC      | forward error correction                    | ARQ      | automatic repeat request                   |
| BCH      | Bose–Chaudhuri–Hocquenghem code             | RS       | Reed–Solomon code                          |
| TCM      | trellis coded modulation                    | PN       | pseudo-noise                               |
| attr(s)  | attribute(s)                                | no.      | number                                     |
| freq     | frequency                                   | info     | information                                |
| ops      | operations                                  | config   | configuration                              |
| rm       | remove                                      | vs, Vs   | versus                                     |
| w/, w/o  | with, without                               | Eg       | example / for example                      |
| +Fig     | draw a figure                               | (I)      | importance / need / significance           |
| →        | leads to / therefore                        | +        | and                                        |

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth  $\geq 6$  marks. Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.